Choose the equation that fits the information

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Constant acceleration / ES: aceleración constante / PT: aceleração constante

Equal velocity changes in equal times.

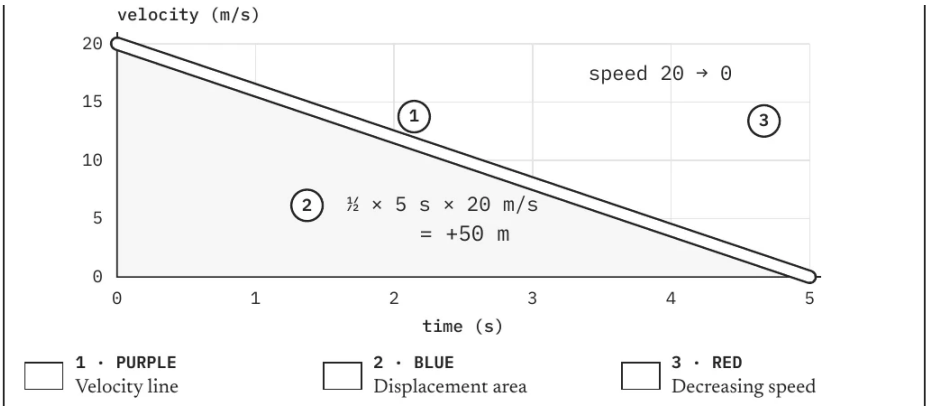
Also on your note page: Known quantities, Displacement, Cross-check.

B READ THE RELATIONSHIP

Choose the equation by the given and missing quantities

As you read, find: given · wanted · constant.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A car moving at +20 m/s brakes at −4 m/s² until it stops. Find its stopping distance, then check the result.

Given $v_i = +20$ m/s $v_f = 0$ m/s $a = -4$ m/s²

Rule No time given: $v_f^2 = v_i^2 + 2a\Delta x$.

Work $0 = 400 - 8\Delta x$ $t = (0 - 20)/(-4)$ area = ½(20)(5)

Answer $\Delta x = +50$ m 5 s to stop; area also gives 50 m .

E SAVE FOR THE FINAL MEETING

For XJ, choose a relationship that fits both the known values and the motion being assumed. A graph provides a second check before a forecast is reported.

Read the physics: Constant acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

List what you know and what you want. A steady velocity needs no acceleration. A steady change in velocity needs a constant-acceleration model.

02 HOW IT WORKS

On a velocity–time graph, slope gives acceleration. The signed area gives displacement. For constant acceleration, the average of the first and last velocities gives the average velocity.

03 WHAT TO CHECK

Choose an equation that uses your known values. Then check the sign and the unit. A matching graph gives a second way to test the result.

04 TRY THE EXAMPLE FROM YOUR NOTES

A car moving at +20 m/s brakes at −4 m/s² until it stops. Find its stopping distance, then check the result. $\Delta x = +50$ m 5 s to stop; area also gives 50 m .

Model check: Choose the equation by the given and missing quantities.

05 CONNECT TO XJ

For XJ, choose a relationship that fits both the known values and the motion being assumed. A graph provides a second check before a forecast is reported.

SAY IT BEFORE YOU WRITE

What must stay constant to average the endpoint velocities? I chose ___ because I know ___ and need ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Constant acceleration

ES: aceleración constante / PT: aceleração constante

Meaning in my words	How can you tell?
How would you explain constant acceleration to a friend? In my words, this means ___.	The velocities are 1, 3 and 5 m/s one second apart. What repeats? I can tell because ___.
Example or sketch	**Non-example and why**
If those gains continue, what velocity would you write for the next second? My example shows ___.	Someone adds 2 m to position every second and calls that speeding up. What pattern did they draw? I would change ___ because ___.

Known quantities

ES: magnitudes conocidas / PT: grandezas conhecidas

Meaning in my words	How can you tell?
What do we mean by the known quantities in a problem? In my words, this means ___.	A cart starts at rest and is timed for 3 s. Which two values can you write down? I can tell because ___.
Example or sketch	**Non-example and why**
A sensor reads +4 m at the start. How would you enter that known value? My example shows ___.	A friend lists a guessed final speed among the given values. Where should it go? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Displacement

ES: desplazamiento / PT: deslocamento

Meaning in my words	How can you tell?
How would you explain displacement to a friend? In my words, this means ___.	Which two marks tell you the displacement even if the cart turns around? I can tell because ___.
Example or sketch	**Non-example and why**
Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement? My example shows ___.	Someone adds both legs and gets 6 m. What did they find? I would change ___ because ___.

Cross-check

ES: comprobación independiente / PT: verificação independente

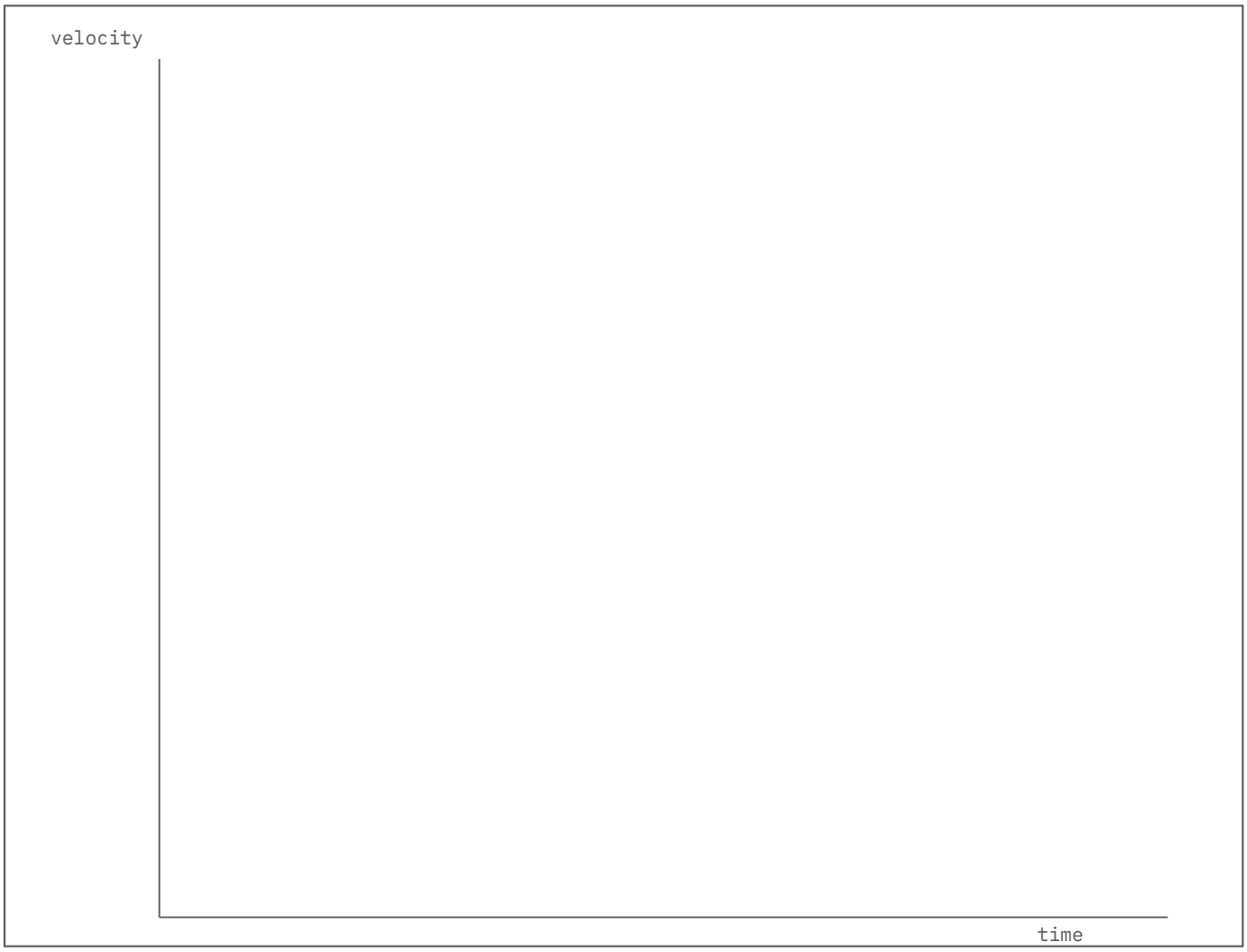
Meaning in my words	How can you tell?
How would you explain cross-check to a friend? In my words, this means ___.	You found displacement using an equation. What feature of a velocity–time graph could check it? I can tell because ___.
Example or sketch	**Non-example and why**
A cart stays at +3 m/s for 2 s. How can a graph check the result of 6 m? My example shows ___.	Someone copies the same number twice. What could actually catch a wrong answer? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a straight velocity–time line. Label which feature gives acceleration and which gives displacement.

Labels you can use: given · wanted · constant.



Tell a partner

I chose ___ because I know ___ and need ___.

Catch the next shot

08

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity-time graph to predict displacement.

CHOOSE ONE JOB TOGETHER

CAMERA ENGINEER ☐

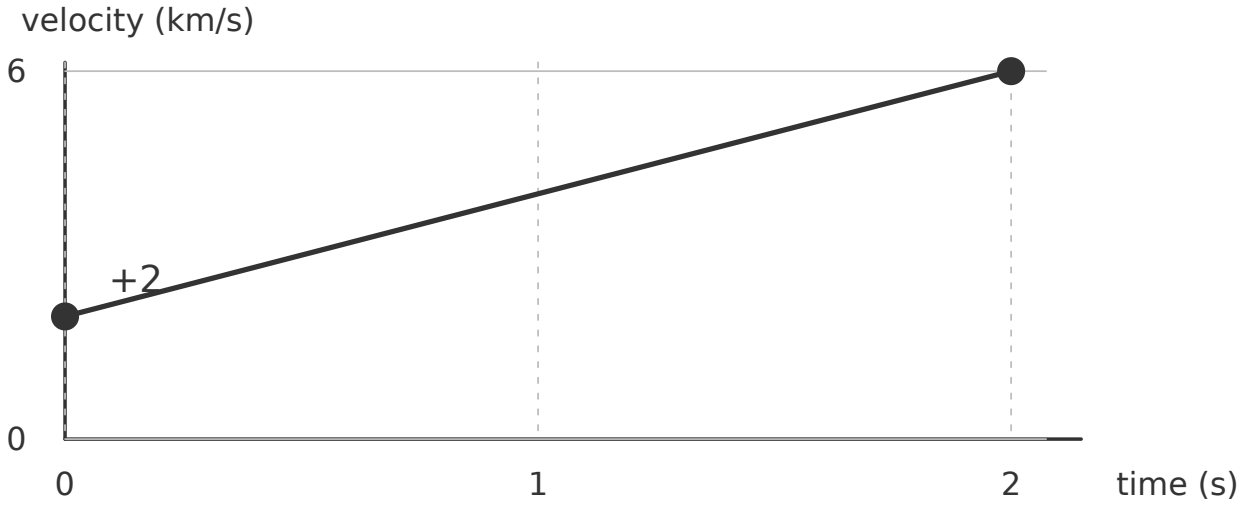
Shade the area under the velocity-time trace. Send the camera's displacement at the end of the shot.

REPLAY EDITOR ☐

Explain why using the final speed for the whole clip would miss the shot. Point to the graph as you speak.

Word help: area / constant / displacement

XJ moves from **+2** to **+6 km/s** over **2 s**. The rehearsal assumes constant acceleration. Where should the next camera look relative to the start?



01 OUR FIRST CALL

Where should the camera look relative to the start? Show how the graph helps.

Our forecast is ___ if ___ stays constant.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new report still let you guarantee that displacement? Explain your revised call.

Now we would say ___ because ___.

Switch speakers. Check your first answer with a kinematics equation. What does that equation treat as constant?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks